

Demonstrating Trusscillator: a System for Fabricating Human-Scale Human-Powered Oscillating Devices

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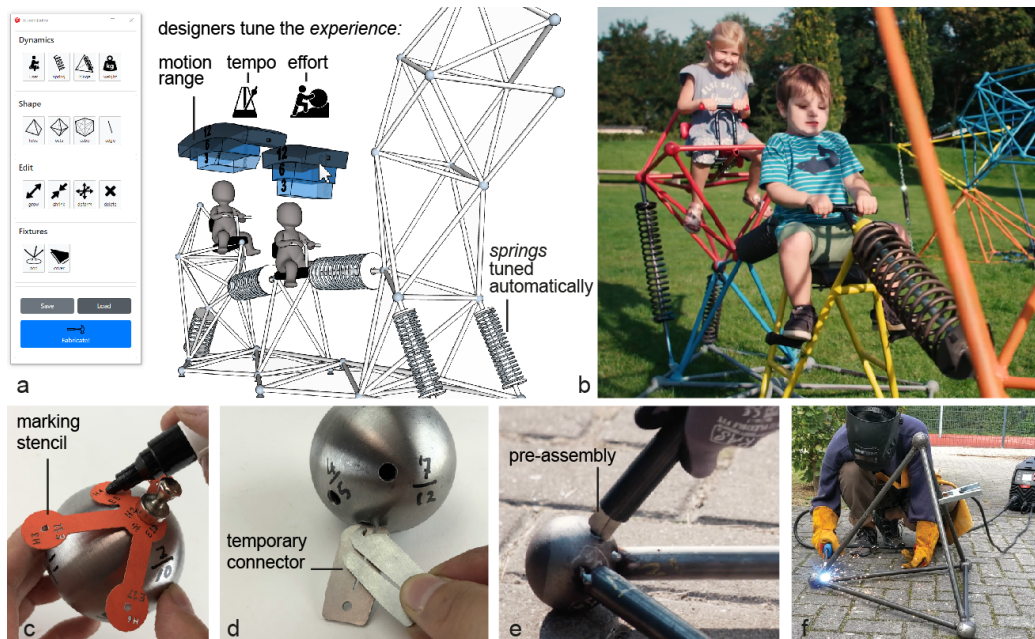


Figure 1: (a) Trusscillator is an integrated system that allows users to design human-scale, human-powered machines, such as this dinosaur-inspired playground device. Trusscillator’s user interface allows designers to interactively construct a steel truss structure, add coil springs, and specify the requirements in terms of motion range, speed, and physical effort. Trusscillator responds by adjusting the coil springs and adding mass so as to produce the desired behavior. (b) The resulting interactive dinosaur sculpture designed for two children challenges the riders to synchronize their movement to causes the sculpture’s head to wiggle. (c) Trusscillator supports steel truss fabrication by generating stencils that show where to attach (d) temporary connectors, (e) that hold steel rods in place, for (f) welding.

ABSTRACT

Trusscillator is an end-to-end system that allows non-engineers to create human-scale human-powered devices that perform oscillatory movements, such as playground equipment, workout devices, and interactive kinetic installations. While recent research has been focusing on generating mechanisms that produce specific movement-path, without considering the required energy for the motion (kinematic approach), Trusscillator supports users in designing mechanisms that recycle energy in the system in the form of *oscillating* mechanisms (dynamic approach), specifically with the help of coil-springs. The presented system features a novel set of tools tailored for designing the *dynamic experience* of the motion. These tools allow designers to focus on user experience-specific aspects, such as motion range, tempo, and effort while abstracting away the underlying technicalities of eigenfrequencies, spring constants, and energy. Since the forces involved in the resulting devices can be high, Trusscillator helps users to fabricate from steel by picking out appropriate steel springs, generating part lists, and producing stencils and welding jigs that help weld with precision. To validate our system, we designed, built, and tested a series of unique playground equipment featuring 2-4 degrees of movement.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.; • **Applied Computing** → Operations Research; Computer Aided Manufacturing..

KEYWORDS

personal fabrication, dynamics, mechanical oscillation, welding

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1 INTRODUCTION

The related work in personal fabrication [1] offers numerous examples of so-called *kinematic systems* [6] that allow users to design and fabricate mechanisms that perform user-specified movement patterns. Examples include the 3D-printed pantograph from *Meta-material Mechanisms* [3], the 5m-tall dinosaur from *TrussFormer* [5], and the animated cheetah created from *Computational Design of Mechanical Characters* [2].

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In this work, we want to extend this line of work towards machines that are *human-powered*, such as playground equipment, workout devices, and certain types of kinetic installations. “Human-powered” means that these devices need to be operated with the limited power that a human or, in some cases, a child can produce.

Unfortunately, when it comes to designing devices for which *limited power* plays a central role, the aforementioned systems for designing kinematic machines are of little help.

To address this key aspect of the design process, we present Trusscillator, a software system that enables users to create human-scale, *human-powered* machines, such as the playground equipment shown in Figure 1. Trusscillator achieves this by allowing users to add *springs* to their designs. Springs have the ability to transform movement (kinetic energy) into compression (potential energy) and transform that back into movement. Consequently, springs help to keep these devices in motion with little effort and thus allow even larger machines to be *human-powered*.

To allow designers to create human-powered movement, Trusscillator offers a novel set of tools, specifically designed for creating oscillating-motion experiences (Figure 1a). These tools allow designers to focus on user experience-specific aspects, such as *motion range*, *tempo*, and *effort* while abstracting away the underlying technicalities of eigenfrequencies, spring constants, masses, and energy use. Because of the potentially large forces, devices designed using Trusscillator are made *from steel* (Figure 1c-f). Trusscillator helps users to fabricate these devices not only by picking out appropriate springs but also by producing stencils and placing temporary connectors that help *welding* the resulting large-scale structures.

2 TRUSSCILLATOR SYSTEM

Trusscillator’s workflow consists of two main phases: (1) the design phase, and the (2) fabrication phase.

2.1 Designing with Trusscillator

As shown in Figure 2a, the playground designers start with creating the frame of the swing by stacking truss-primitives, specifically tetrahedra, and octahedra and inserting springs-telescopes (building on *TrussFab* [4] and *TrussFormer* [5]). They place a *ragdoll figure* onto the model, which inserts a matching seat for a child.

As illustrated by Figure 2b, Trusscillator displays the properties of this basic swing using what we call the *motion bar*: an average 6-year-old should be capable of making it rock roughly by the amplitude indicated by the middle curved blue bar labeled “6”. Designers can play back a simulation of the child rocking by clicking on this bar.

To fine-tune the experience, designers adjust the device’s amplitude by grabbing the handle attached to the *motion bar* and drag it inwards. Trusscillator responds by re-running its optimization engine and replaces the springs in the model with springs that produce motion in the request range.

The reduction in amplitude required stiffer springs, that now caused the ride to oscillate faster (0.6s period, indicated on hover). As shown in Figure 3a, Trusscillator considers this uncomfortable and displays a notification (in the shape of a metronome, together with the word “fast”). The designers click the notification to change

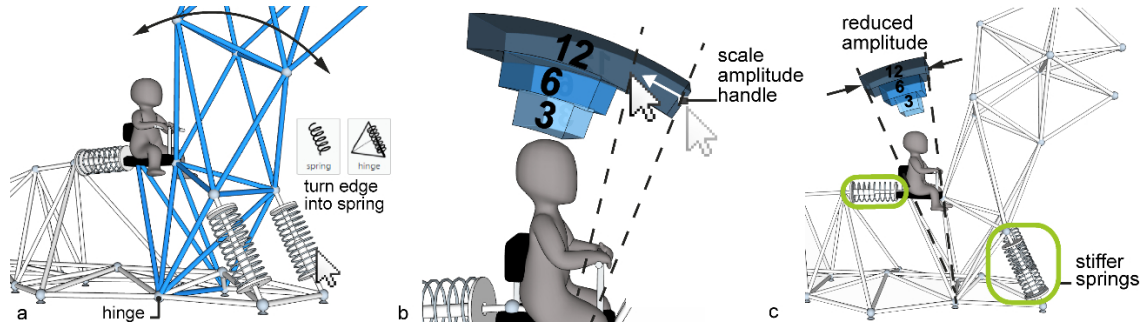


Figure 2: (a) Using the spring tool, designers enable parts of the model to move. The newly created moving part of the model gets briefly highlighted in blue. (b) The resulting oscillating motion is summarized in form of a motion-bar above the user, calculated for multiple age groups. When designers adjust the motion space by dragging the scale handle, (c) Trusscillator finds a combination of softer springs that will produce the requested amplitude.

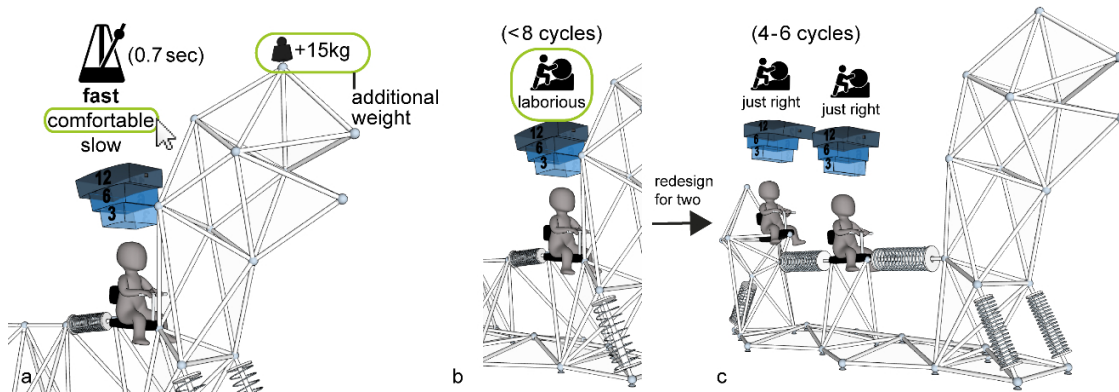


Figure 3: (a) When designers changing the tempo widget from slow to comfortable Trusscillator runs its optimization and adds additional weight to the tip of the head to reduce the resonant frequency and tunes the springs again to maintain amplitude. (b) Reducing the effort would require cutting on the weight of the structure, which designers can't do. (c) Instead, they add one more seating position. The final design comprises three spring-coupled inverted pendula, the head, middle seat, and tail.

it to “comfortable”. Trusscillator responds by re-running its optimization to find a frequency in the range that is considered a pleasant rocking period (0.8-1.2s), which it achieves by making yet another adjustment to the springs, as well as by adding a weight to the head of the dinosaur as shown in Figure 3b.

At this point, designers notice a third concern: the *effort* widget suggests “laborious” (Figure 3a). This means the device requires more than 8 cycles to reach maximum amplitude, bearing the risk of children losing interest before getting it into full swing. One of the designers proposes clicking the effort widget to reduce the effort (see section Figure 3), but the other designer sees the opportunity to add another level of excitement and challenge to the design by bringing in a second child. As illustrated by Figure 3c, they add a second seat and yet another spring.

This update changes the widget from *laborious* to *just right* for both children, as now they both contribute power. More importantly, the resulting device has now created an additional challenge—a social challenge: first, it requires the first child to recruit another child as confederate to produce in order to successfully get the device to reach peak amplitude. Second, it requires the two children

to synchronize their movement (or to decide to play against each other). Trusscillator allows for this by running its optimization procedure to tune the two seats to similar eigenfrequencies. To get a sense of what the resulting synchronization will feel like, the designers invoke simulations of the resulting movement (by clicking on the motion bars for each of the three age groups).

The designers satisfied with the result, hit the *export & fabricate* button and proceed with fabrication of their device.

2.2 Fabrication pipeline

Trusscillator now prepares the necessary steps to fabricate the designed structures from steel rods, steel spheres, and steel springs, which users assemble using a power drill, an angle grinder, and a welding device (Figure 1c-f).

As a first step, Trusscillator generates a list with the lengths of required steel tubes, the number of steel balls to be purchased, and a list of the steel springs to be purchased.

The main challenge in welding truss structures is precisely pre-aligning all components in space, as only a slight deviation in one angle will cause non-fitting edges. Trusscillator supports the user

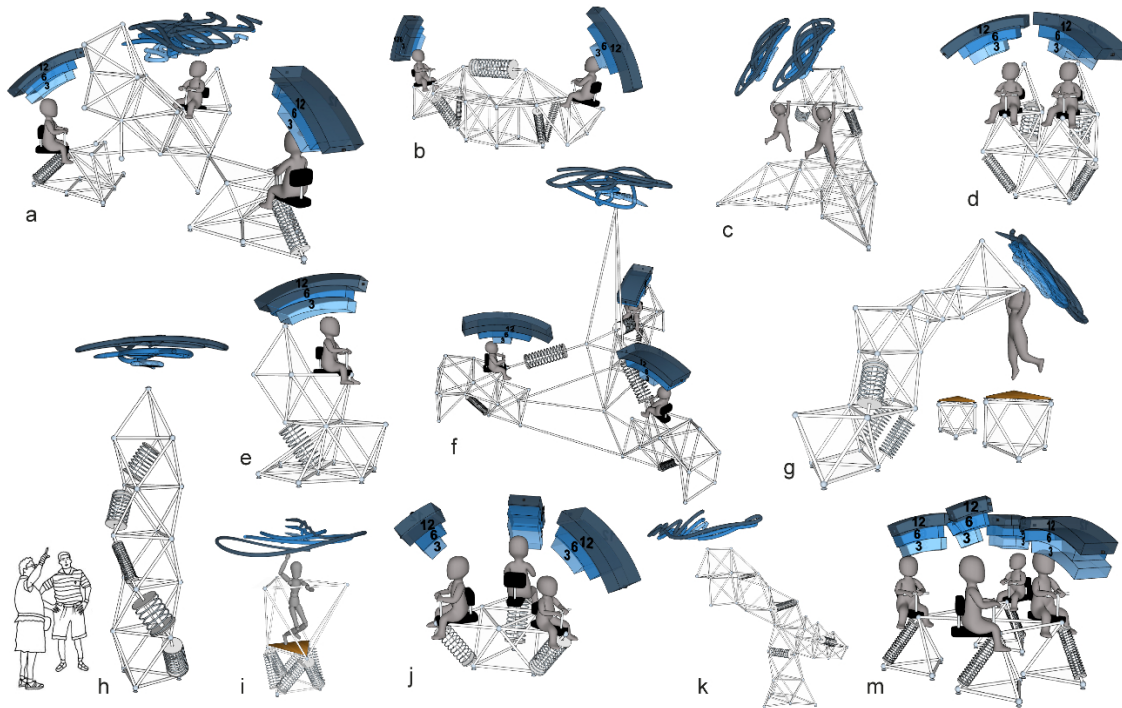


Figure 4: Some of the designs we created using Trusscillator.

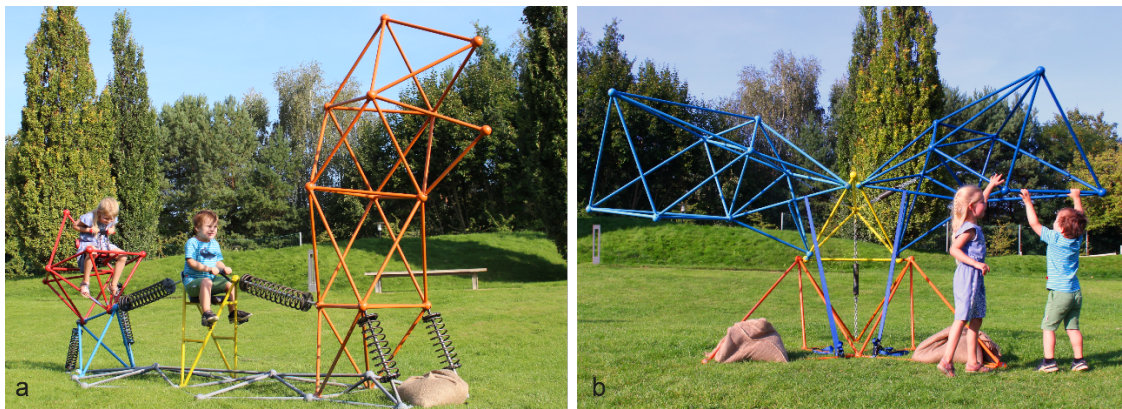


Figure 5: (a) In our dinosaur-shaped swing children can induce resonance by synchronizing their motion. (b) The “bird-swing” structure was designed to allow children to swing in 3D space and also to be able to influence each other’s experience. Both models were fabricated in 2-3 days, supported end-to-end by Trusscillator.

by generating stencils for marking the connection points on the nodes-spheres (Figure 1c). Using our temporary connection system (Figure 1d) users set up the provisional structure (Figure 1e), and finally weld the entire structure (Figure 1f).

3 DESIGN SPACE & VALIDATION

We have used Trusscillator to design a wide range of devices. The samples are shown in Figure 4 including swings featuring 1D (b, e, j, m), and 3D motion (a, c, f, g), as well as kinetic installations (h, k) and balancing workout equipment (i).

To validate Trusscillator’s functionalities, two of the models were fabricated physically, i.e., the “brachiosaurus” from Figure 1, and our “bird swing” design, as shown in Figure 5. Trusscillator allowed a team of two to design, cut, drill, assemble, weld, and paint each model in 2-3 days.

REFERENCES

- [1] Patrick Baudisch and Stefanie Mueller. 2017. Personal Fabrication. *Foundations and Trends in Human-Computer Interaction* 10, no. 3–4 (2017): 165-293.

- [2] Stelian Coros, Bernhard Thomaszewski, Gioacchino Noris, Shinjiro Sueda, Moira Forberg, Robert W. Sumner, Wojciech Matusik, and Bernd Bickel. 2013. Computational design of mechanical characters. *ACM Trans. Graph.* 32, 4, Article 83 (July 2013), 12 pages. DOI:<https://doi.org/10.1145/2461912.2461953>
- [3] Alexandra Ion, Johannes Frohnhofen, Ludwig Wall, Robert Kovacs, Mirela Alistar, Jack Lindsay, Pedro Lopes, Hsiang-Ting Chen, and Patrick Baudisch. 2016. Meta-material Mechanisms. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST '16)*. Association for Computing Machinery, New York, NY, USA, 529–539. DOI:<https://doi.org/10.1145/2984511.2984540>
- [4] Robert Kovacs, Anna Seufert, Ludwig Wall, Hsiang-Ting Chen, Florian Meinel, Willi Müller, Sijing You, Maximilian Brehm, Jonathan Striebel, Yannis Kommana, Alexander Popiak, Thomas Bläsius, and Patrick Baudisch. 2017. TrussFab: Fabricating Sturdy Large-Scale Structures on Desktop 3D Printers. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. ACM, New York, NY, USA, 2606–2616. DOI: <https://doi.org/10.1145/3025453.3026016>
- [5] Robert Kovacs, Alexandra Ion, Pedro Lopes, Tim Oesterreich, Johannes Filter, Philipp Otto, Tobias Arndt, Nico Ring, Melvin Witte, Anton Synytsia, and Patrick Baudisch. 2018. TrussFormer: 3D Printing Large Kinetic Structures. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (UIST '18)*. Association for Computing Machinery, New York, NY, USA, 113–125. DOI: <https://doi.org/10.1145/3242587.3242607>
- [6] Asok Kumar Mallik, Amitabha Gosh, Günter Ditttrich. 1994. Kinematic analysis and synthesis of mechanisms. CRC Press.